# JONATHAN MA

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## Education

**Cornell University, College of Engineering (Ithaca, NY)**

Master of Engineering, Systems Engineering *Anticipated December 2026*

Bachelor of Science, Computer Science, Minor in Robotics (GPA 3.99) *May 2026*

- Coursework: Systems Programming, OS, Intro to Robotics, Computer Vision, Computer Graphics, AI/ML
- TA for Computer Systems (C, RISC V Assembly), OCaml Programming, and Intro to Robotics (ROS) classes
- Honors: Dean's List, USRA National Aerospace Scholarship, Interstellar Research Group Scholarship

## Experience

**NVIDIA: Embedded Software Engineering Intern** *May - August 2026*

- Developed firmware for OpenBMC out-of-band baseboard management controller platform for data centers
- Revamped multi-platform hardware acceptance test suite with 800+ tests to ensure maintainability/scalability
- Designed modular/extensible emulation-simulation platform to allow firmware validation without hardware

**SpaceX: Flight Software Engineering Intern (Starshield)** *May - August 2025*

- Architected C++ flight and ground software solution spanning 5 satellite systems to deconflict satellite ops
- Conducted over 20 rigorous hardware-in-the-loop tests to ensure mission readiness with embedded Linux
- Organized design review and collaborated with 10+ stakeholders across teams, converting ambiguous guidelines to actionable tasks in order to develop a robust solution and testing plan
- Overhauled satellite integration and testing Python script to increase modularity and testability, taking ownership to drastically reduce risk for critical path test campaign while saving 1000s of engineering hours

**Cornell Space Systems Design Studio: Alpha CubeSat Flight Software Lead** *Oct 2022 - May 2026*

- Promoted to lead of 5 person team after 2 yrs for exemplary performance and initiative to learn/solve problems
- Engineered real-time embedded C++ flight software for 1U CubeSat & light sail payload on 6 mo LEO mission
- Developed code for memory constrained microcontroller with 32KB flash to interface with avionics firmware/drivers and control GPS, IMU, and LoRa radio peripherals via SPI, I2C, and UART protocols
- Built and deployed full-stack ground station on Linux system with intuitive React and JavaScript control UI, Python Flask backend API, and Nginx reverse proxy to handle 100+ command & telemetry data points

**RTX Collins Aerospace: Systems Engineering Intern (SEPP)** *May - August 2024*

- Collaborated with team of interns to conduct computer networking research studies using Info Centric Routing over IP with Raspberry Pi fleet, and extended 50K line open source C++ library to support network persistency

**Johns Hopkins University Applied Physics Lab: Ground Software Engineering Intern** *May - August 2023*

- Led team of interns through design and creation of Angular and Java based app for parsing binary spacecraft command and telemetry packets for NASA's IMAP and Dragonfly missions, used by 50+ employees
- Applied Agile methodologies and solicited feedback from project leads throughout project duration

**First Tech Challenge Robotics Team #7303: Robot Automation Lead** *Aug 2019 - June 2022*

- Won Maryland Tech Invite out of top 32 teams globally, Control Award for most innovative control/automation
- Collaborated w/ team to develop OpenCV perception, odometry localization, motion planning & PID control
- Developed multistage automation and CV algorithms which can autonomously generate paths to collect and shoot rings while moving at 5 in target from anywhere on a 12x12 ft field

## Programming Projects

- Birdsong Synthesizer (C), Applied DSP direct digital synthesis algorithm to synthesize birdsongs on ARM-based Raspberry Pi 2040 microcontroller with DAC and ADC using the SPI protocol and GPIO pins
- MirrorBot (C, Python), Developed mirror control algorithm for robot using ROS, Arduino & Intel RealSense depth camera to detect faces and dynamically adjust reflections to encourage interactions in public spaces
- Rubik's Cube Solving Robot (C, Python), Arduino powered robot optimized to solve Rubik's Cube in 3-4 sec
    - Created OpenCV pipeline to scan cube and create Rubik's cube mosaic, breadboarded Arduino circuit

## Technical Skills

- Languages (Proficient): Python, C++, C, TypeScript, JavaScript, HTML/CSS, Java, OCaml, SQL, Verilog
- Frameworks: Arduino, ROS, OpenCV, Protothreads, NumPy, PyTorch, Pandas, Flask, React, Angular, Firebase
- Tools: Git, Docker, Linux, Bash, PlatformIO, TCP/IP, Elasticsearch, Kibana, Jira, CI/CD, Claude Code, Cursor